Digital Image Processing

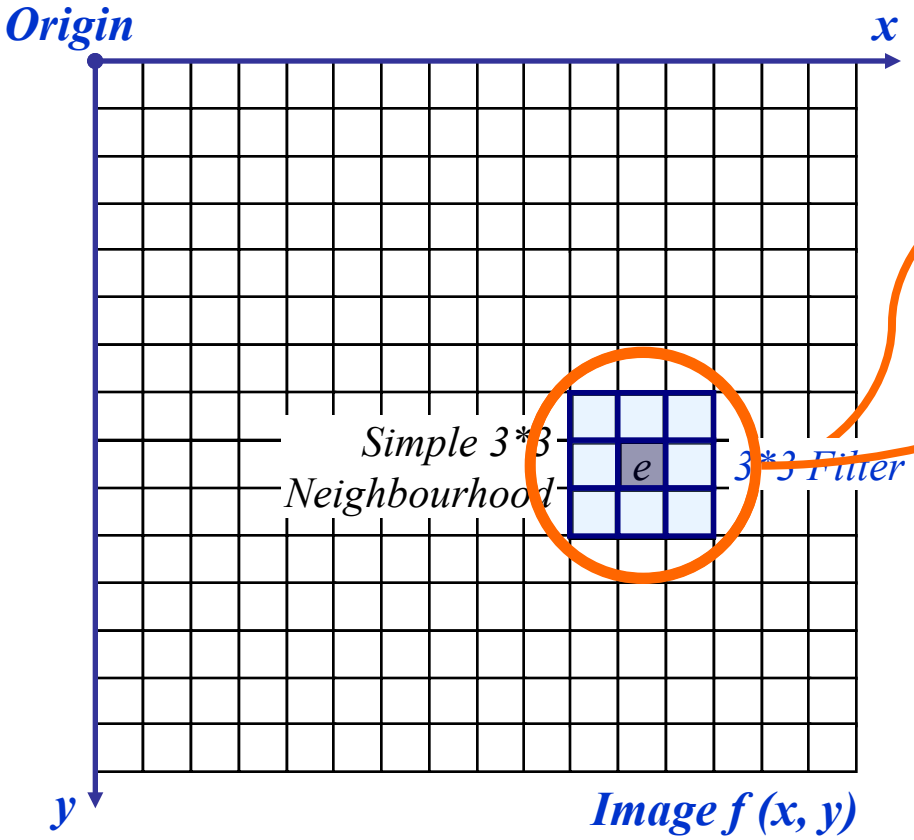
Image Enhancement
(Spatial Filtering 2)

Contents

In this lecture we will look at more spatial filtering techniques

- Spatial filtering refresher
- Sharpening filters
  - 1st derivative filters
  - 2nd derivative filters
- Combining filtering techniques

Spatial Filtering Refresher



a	b	c
d	e	f
g	h	i

Original Image Pixels

*

r	s	t
u	v	w
x	y	z

Filter

$$e_{processed} = v*e + r*a + s*b + t*c + u*d + w*f + x*g + y*h + z*i$$

The above is repeated for every pixel in the original image to generate the smoothed image

3.6 Sharpening Spatial Filters

Previously we have looked at smoothing filters which remove fine detail

Sharpening spatial filters seek to highlight fine detail

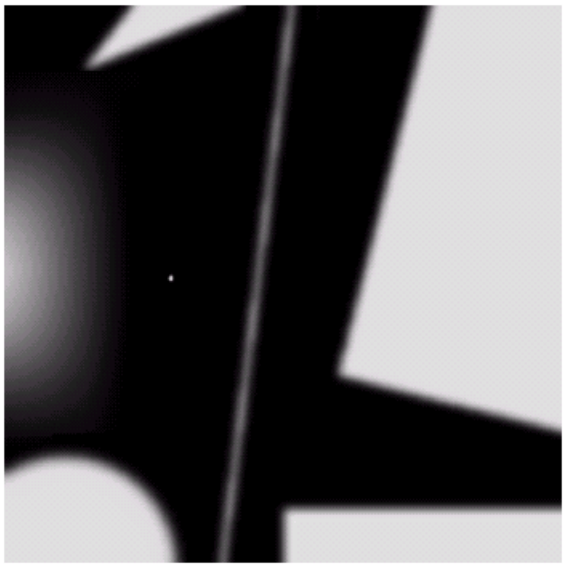
- Remove blurring from images
- Highlight edges

Sharpening filters are based on *spatial differentiation*

Spatial Differentiation

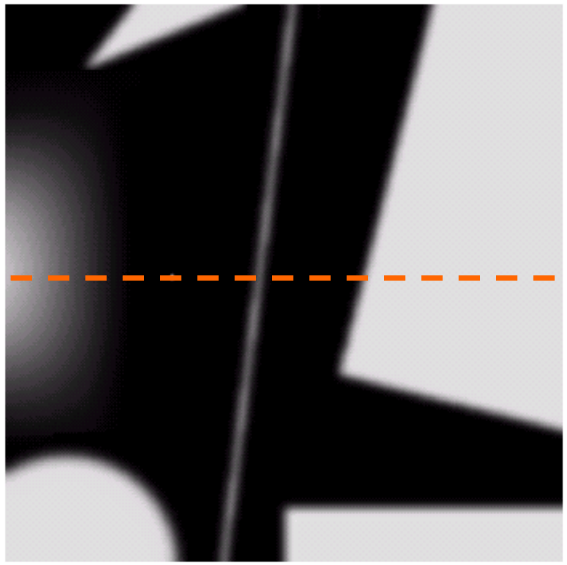
Differentiation measures the *rate of change* of a function

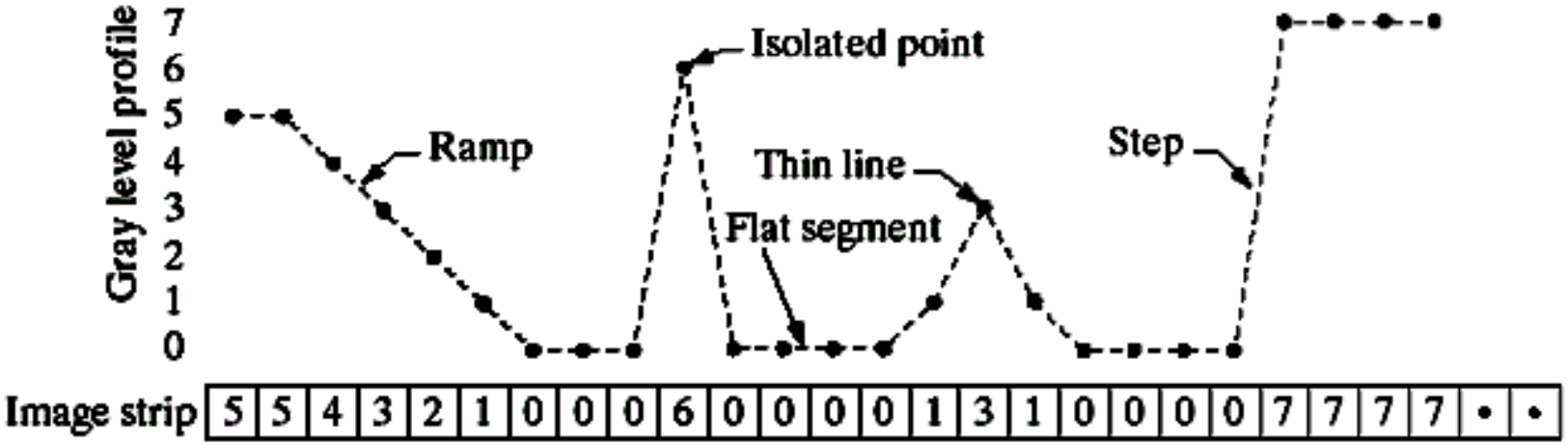
Let's consider a simple 1 dimensional example



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Differentiation





Images taken from Gonzalez & Woods, Digital Image Processing (2002)

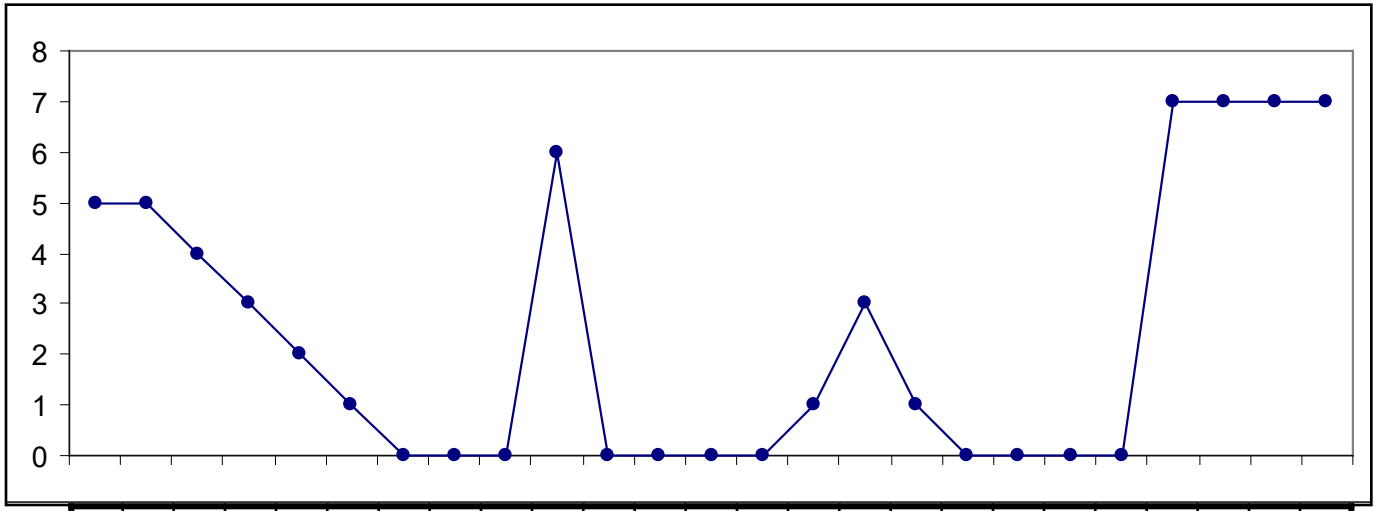
1st Derivative

The formula for the 1st derivative of a function is as follows:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

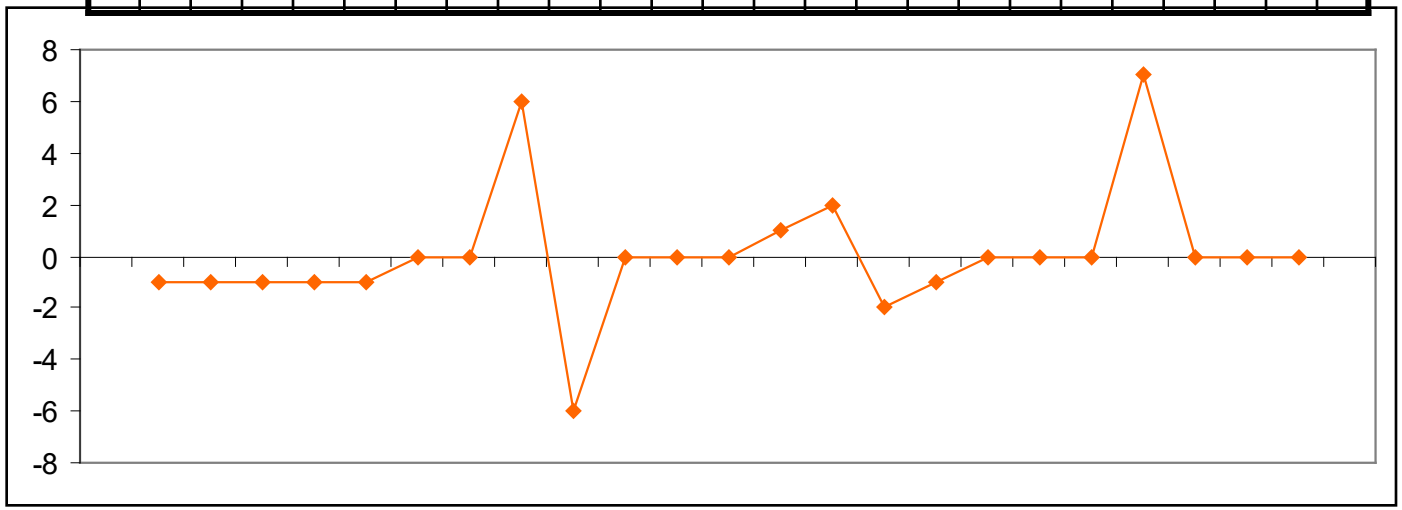
It's just the difference between subsequent values and measures the rate of change of the function

1st Derivative (cont…)



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7

0	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	1	2	-2	-1	0	0	0	7	0	0	0	



2nd Derivative

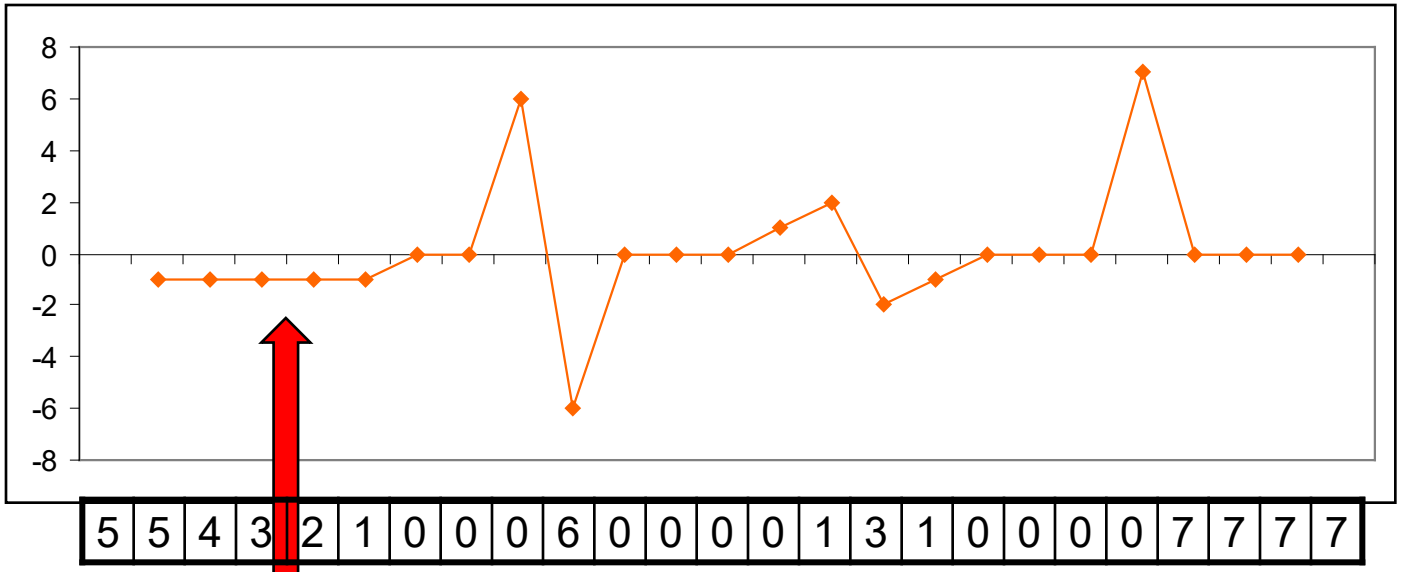
The formula for the 2nd derivative of a function is as follows:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

Simply takes into account the values both before and after the current value

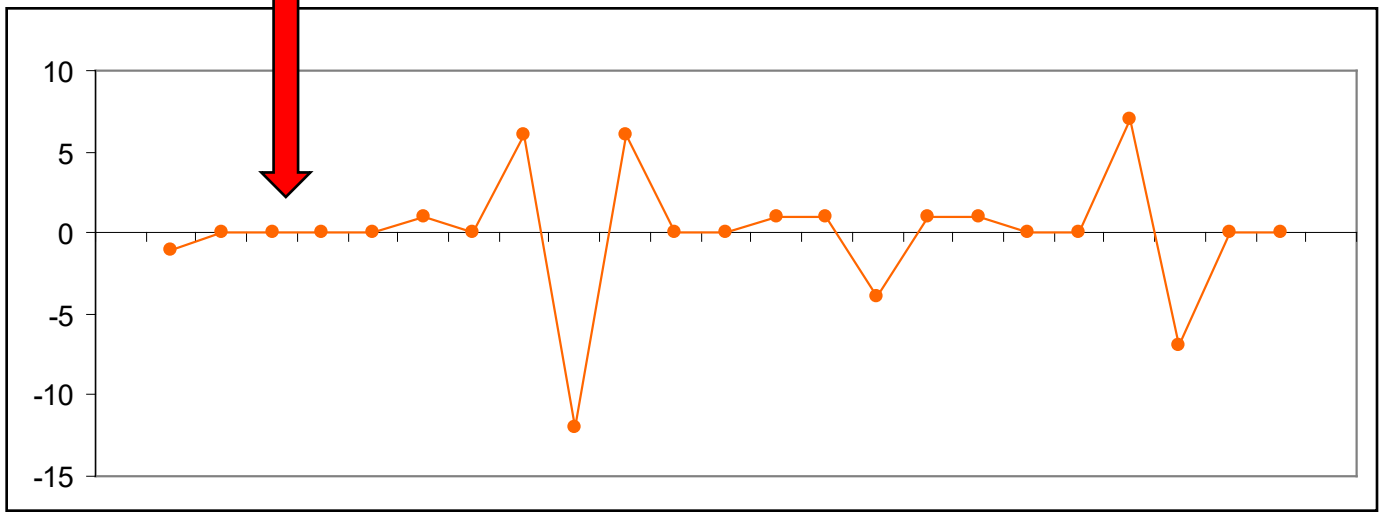
2nd Derivative (cont…)

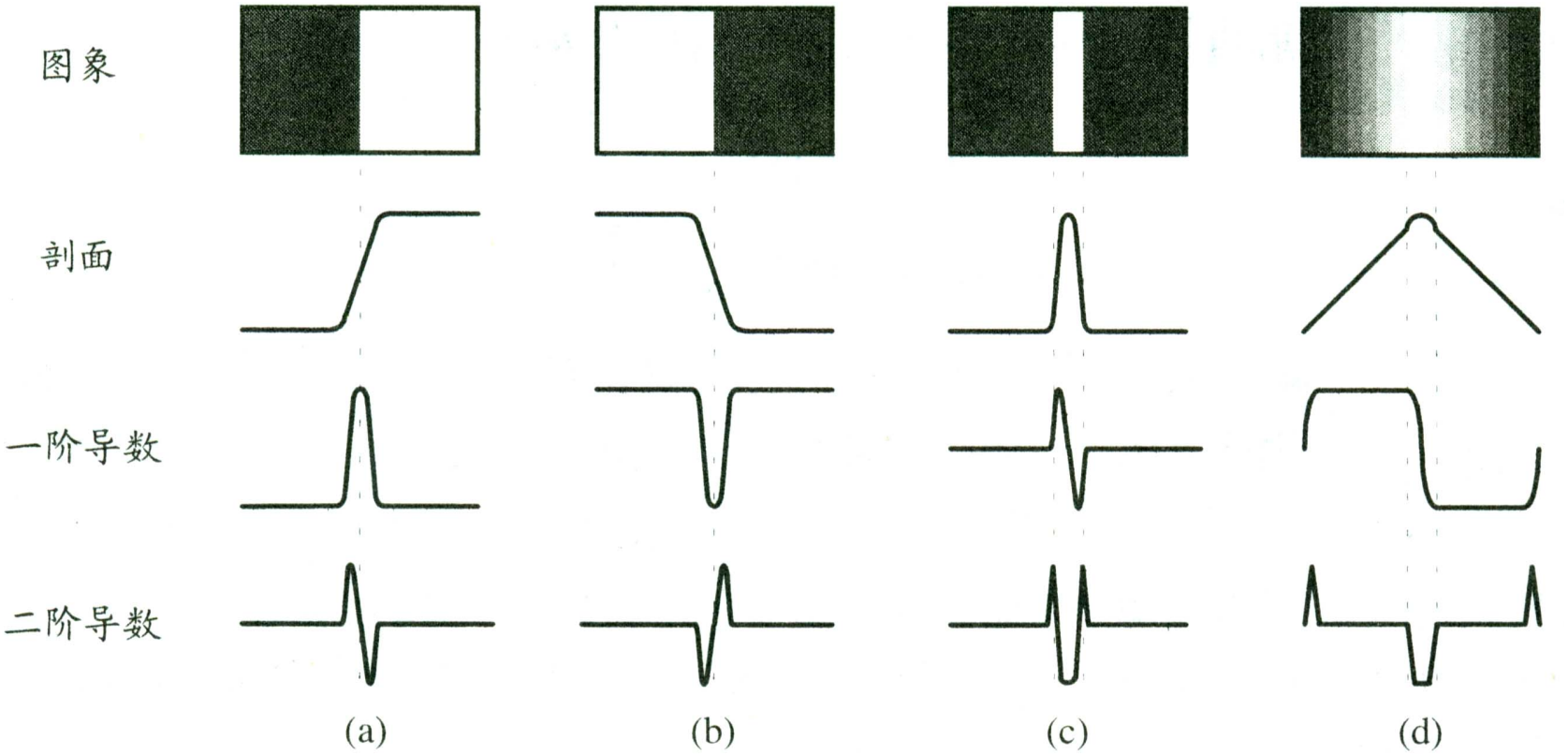
1st



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7

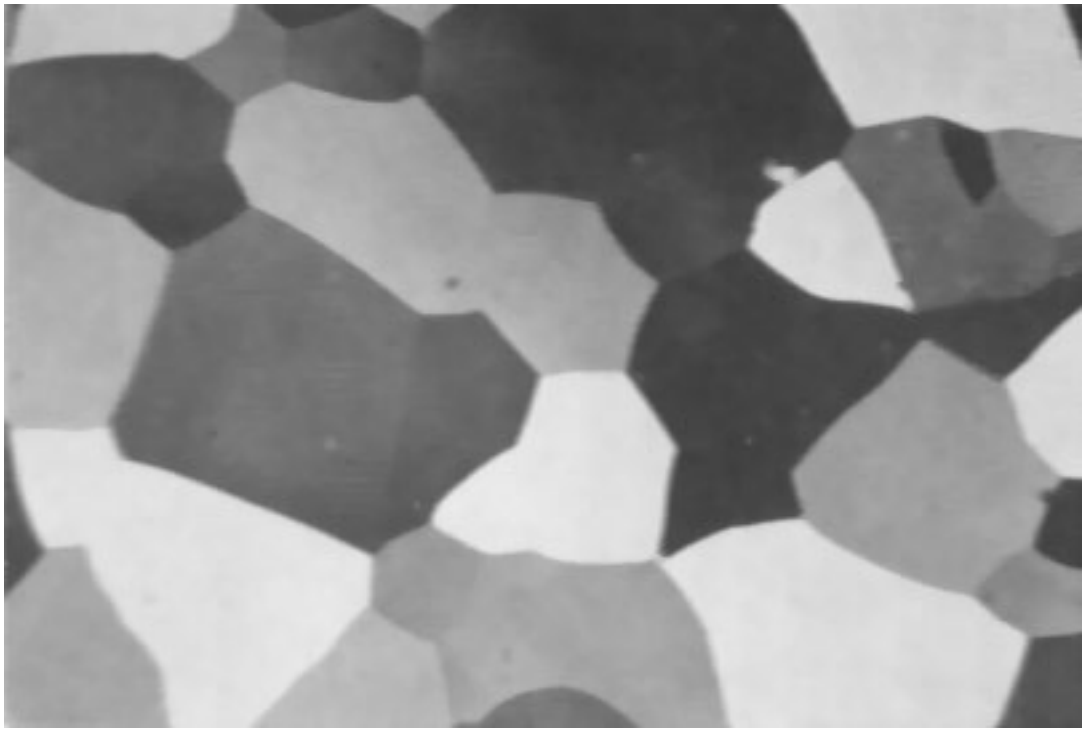
2nd





阶梯状　　阶梯状　　脉冲状　　屋顶状

阶跃边缘



采用一阶微分极大值、二阶微分过零点识别细节

Using Second Derivatives For Image Enhancement

The 2nd derivative is more useful for image enhancement than the 1st derivative

- Stronger response to fine detail

The first sharpening filter we will look at is the *Laplacian*

- One of the simplest sharpening filters
- We will look at a digital implementation

3.6.2 The Laplacian

The Laplacian is defined as follows:

$$\nabla^2 f = \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y}$$

where the partial 2nd order derivative in the x direction is defined as follows:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

and in the y direction as follows:

$$\frac{\partial^2 f}{\partial^2 y} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

The Laplacian (cont…)

So, the Laplacian can be given as follows:

$$\nabla^2 f = [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)] - 4f(x,y)$$

Filter?

We can easily build a filter based on this

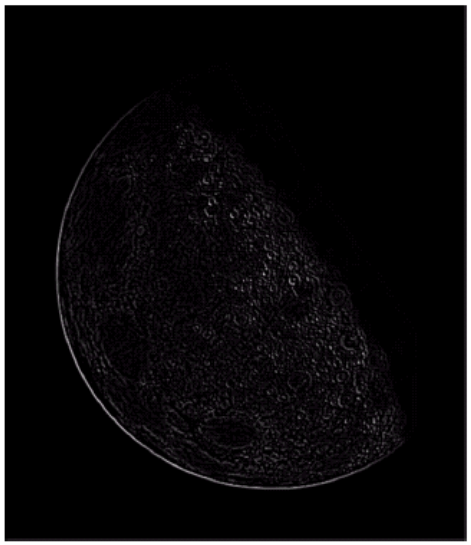
0	1	0
1	-4	1
0	1	0

The Laplacian (cont…)

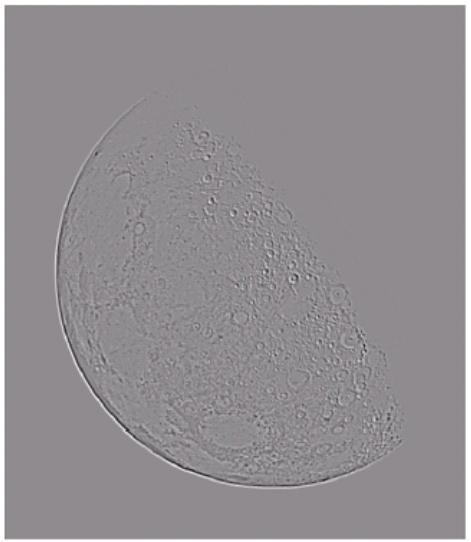
Applying the Laplacian to an image we get a new image that highlights edges and other discontinuities



Original Image



Laplacian Filtered Image



Laplacian Filtered Image Scaled for Display

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

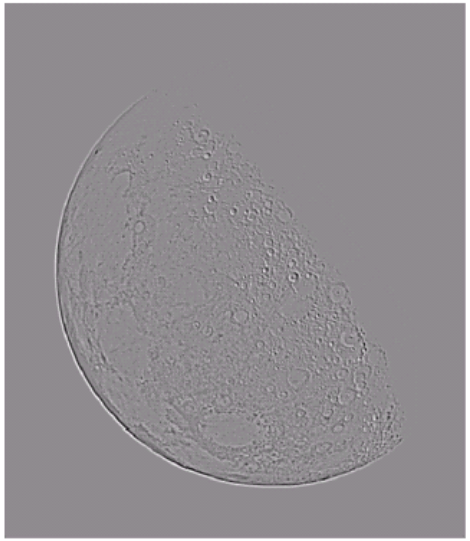
But That Is Not Very Enhanced!

The result of a Laplacian filtering is not an enhanced image

We have to do more work in order to get our final image

Subtract the Laplacian result from the original image to generate our final sharpened enhanced image

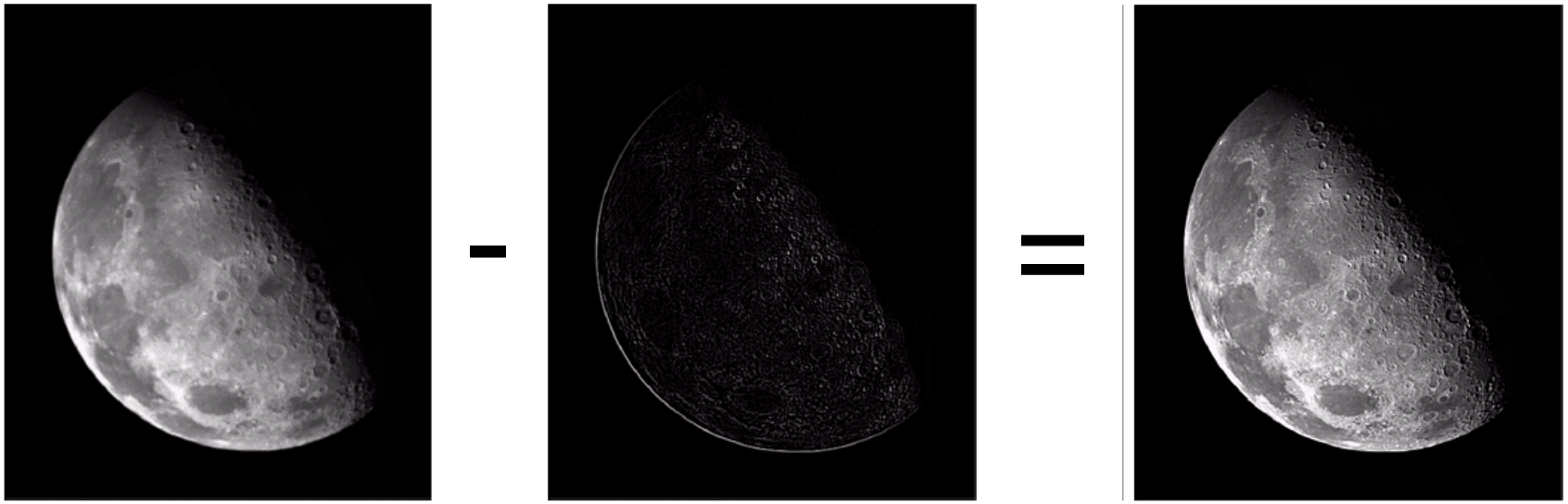
$$g(x,y) = f(x,y) - \nabla^2 f$$



Laplacian
Filtered Image
Scaled for Display

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Laplacian Image Enhancement



Original Image

Laplacian Filtered Image

Sharpened Image

In the final sharpened image edges and fine detail are much more obvious

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Laplacian Image Enhancement





Images taken from Gonzalez & Woods, Digital Image Processing (2002)

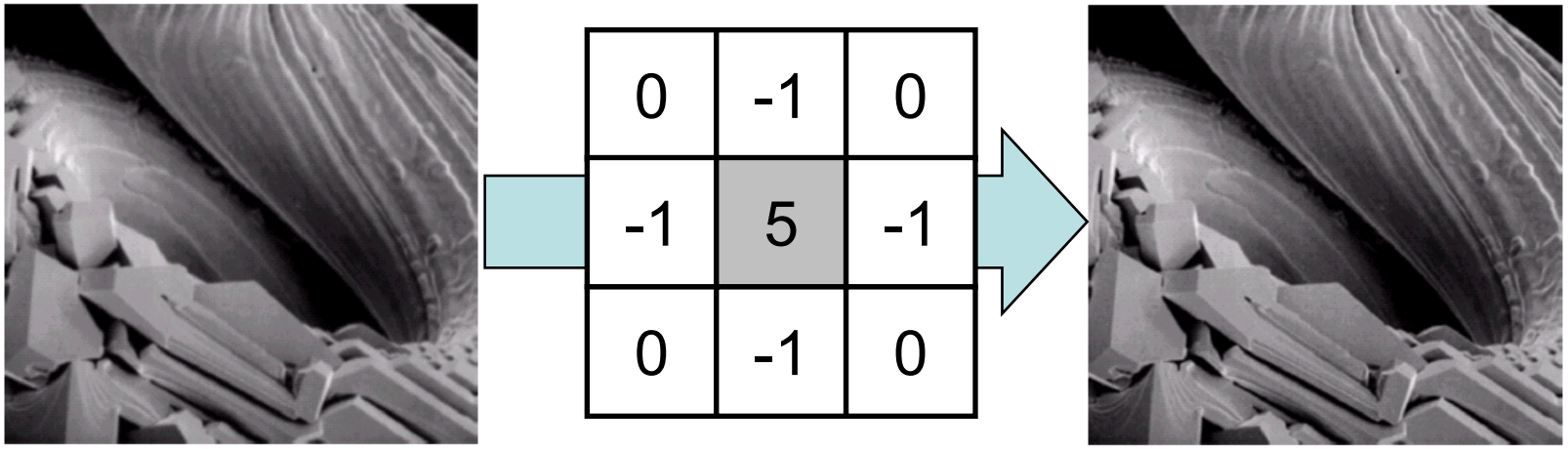
Simplified Image Enhancement

The entire enhancement can be combined into a single filtering operation

$$
\begin{aligned}
g(x,y) &= f(x,y) - \nabla^2 f \\
&= f(x,y) - [f(x+1,y) + f(x-1,y) \\
&\qquad + f(x,y+1) + f(x,y-1) \\
&\qquad - 4f(x,y)] \\
&= 5f(x,y) - f(x+1,y) - f(x-1,y) \\
&\qquad - f(x,y+1) - f(x,y-1)
\end{aligned}
$$

Simplified Image Enhancement (cont…)

This gives us a new filter which does the whole job for us in one step



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Simplified Image Enhancement (cont…)





Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Variants On The Simple Laplacian

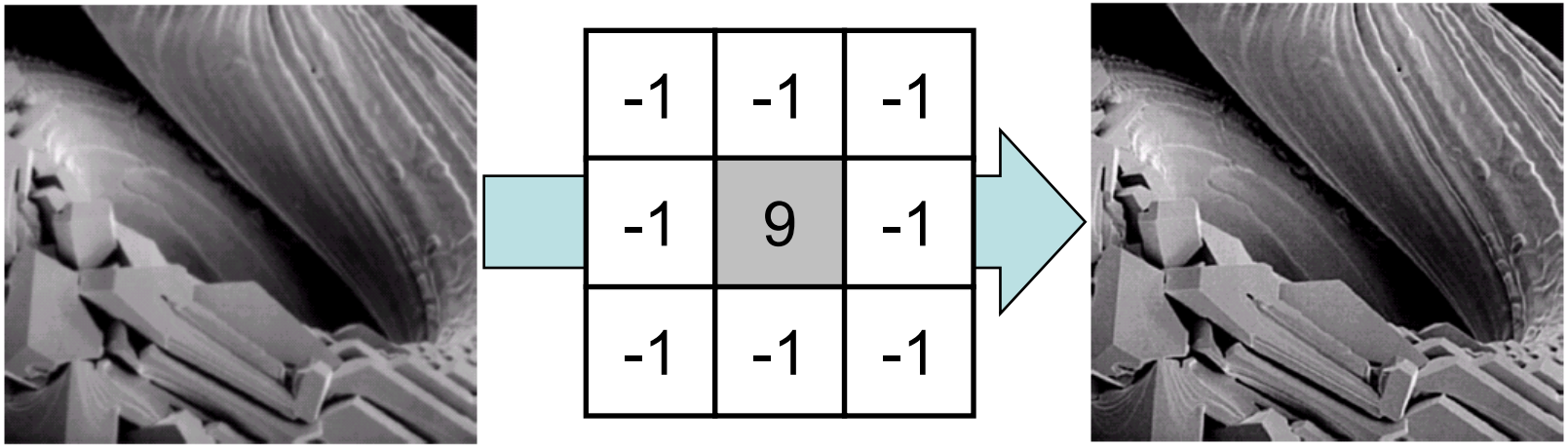
There are lots of slightly different versions of the Laplacian that can be used:

0	1	0
1	-4	1
0	1	0

Simple Laplacian

1	1	1
1	-8	1
1	1	1

Variant of Laplacian



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

问什么能提取高频（边界）？

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{8}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

平滑窗口　　图象锐化模板

图象锐化 —— 相当于从原图象中减去它的低频分量，即原始图象经平滑处理后所得的图象。

3.6.4 1st Derivative Filtering

Implementing 1st derivative filters is difficult in practice

For a function $f(x, y)$ the gradient of f at coordinates (x, y) is given as the column vector:

$$\nabla \mathrm{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

1st Derivative Filtering (cont…)

The magnitude of this vector is given by:

$$\nabla f = mag(\nabla \mathrm{f})$$
$$= \left[G_x^2 + G_y^2\right]^{1/2}$$
$$= \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{1/2}$$

For practical reasons this can be simplified as:

$$\nabla f \approx |G_x| + |G_y|$$

1st Derivative Filtering (cont…)

There is some debate as to how best to calculate these gradients but we will use:

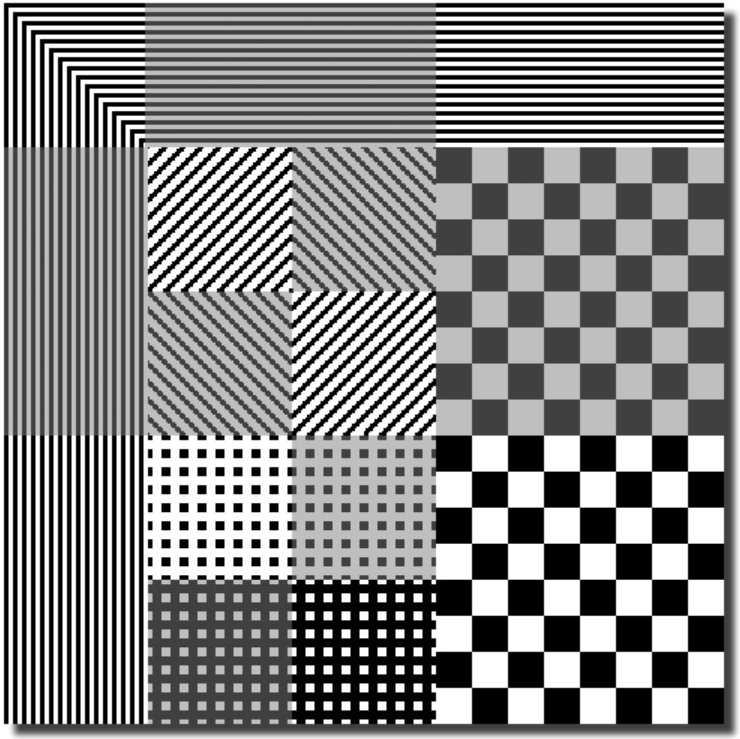
$$\nabla f \approx \left|\left(z_7 + 2z_8 + z_9\right) - \left(z_1 + 2z_2 + z_3\right)\right|$$
$$+ \left|\left(z_3 + 2z_6 + z_9\right) - \left(z_1 + 2z_4 + z_7\right)\right|$$

which is based on these coordinates

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

1st Derivative Filtering (cont…)

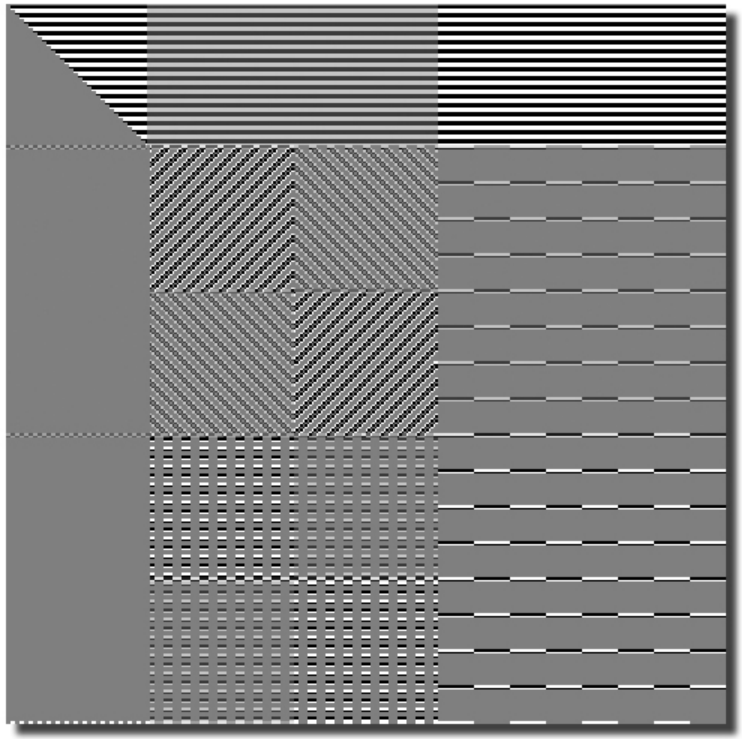
Convolution Examples: Original Images

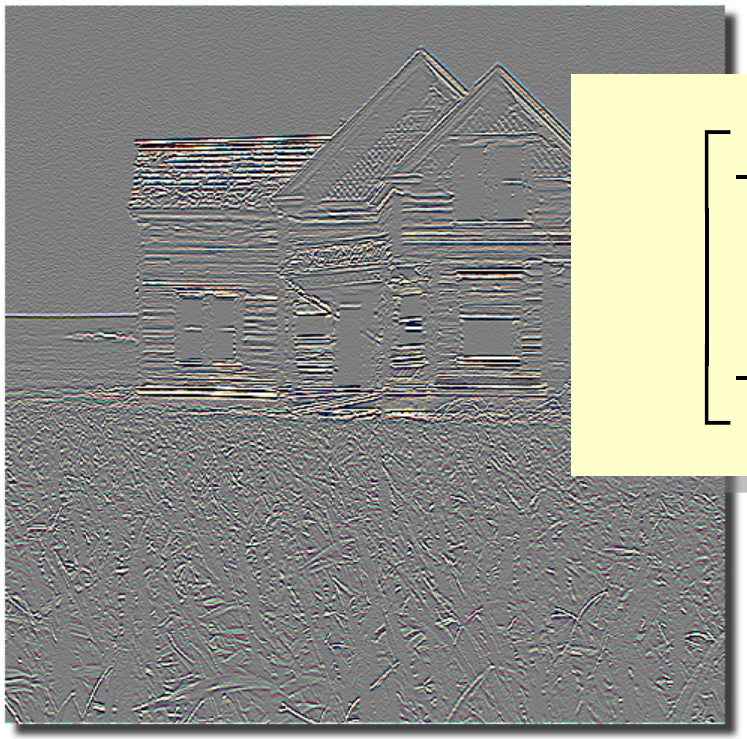




1st Derivative Filtering (cont…)

Convolution Examples: Vertical Difference



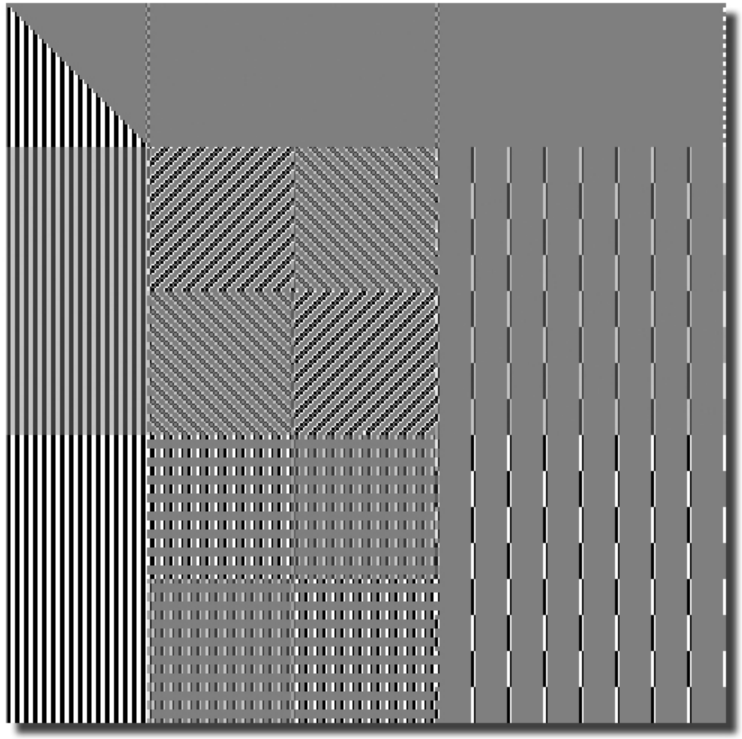


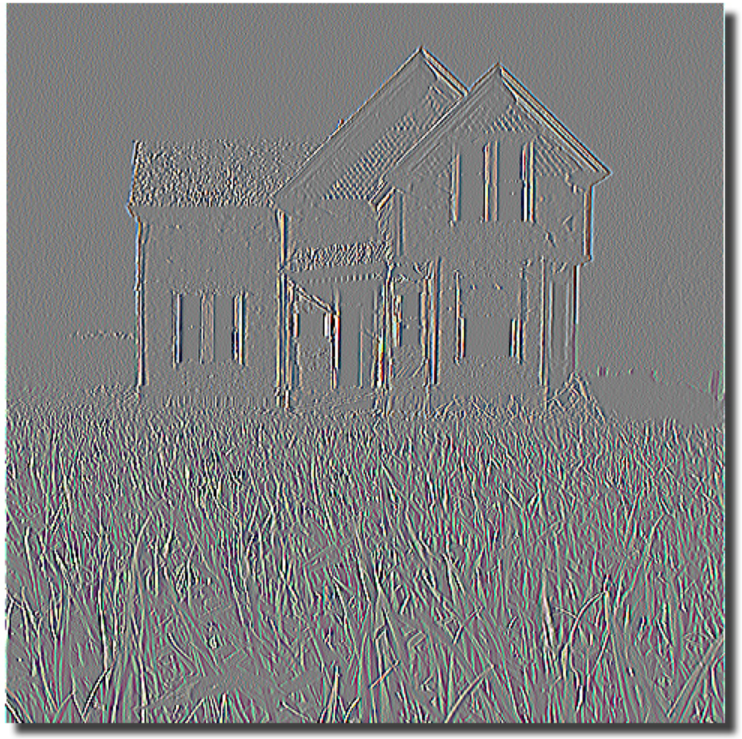
$$\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

1st Derivative Filtering (cont…)

$$\begin{bmatrix} -1 & 2 & -1 \end{bmatrix}$$

Convolution Examples: Horizontal Difference

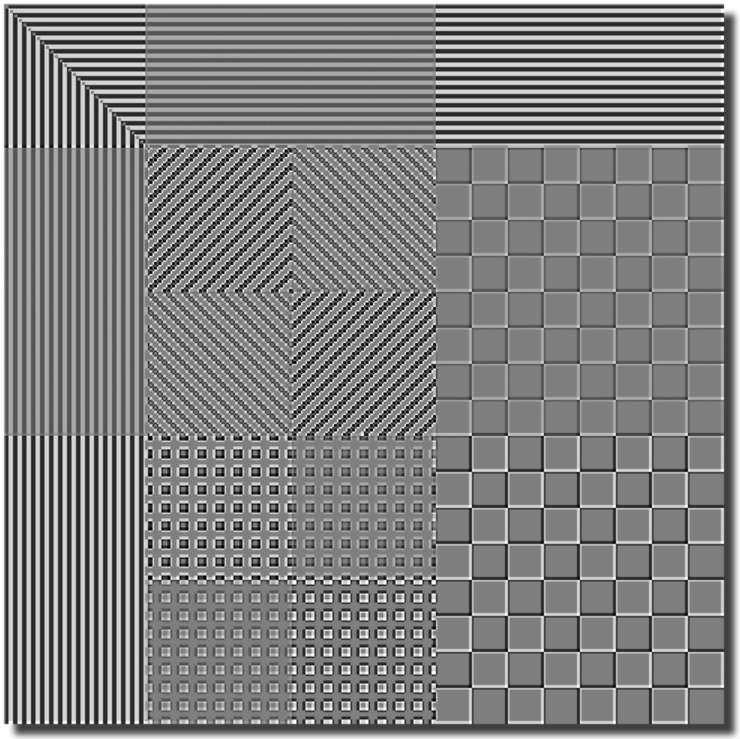




1st Derivative Filtering (cont…)

Convolution Examples: H + V Diff.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

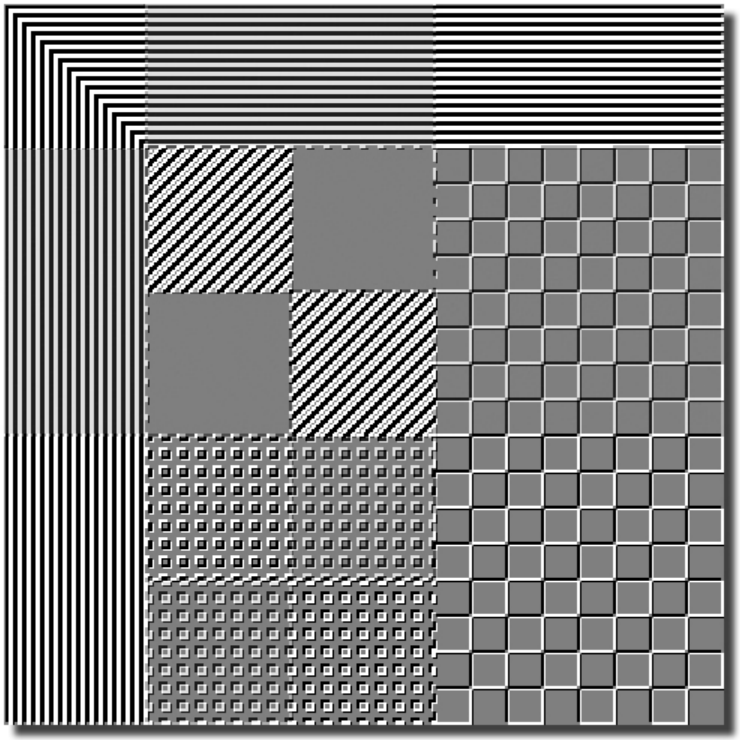


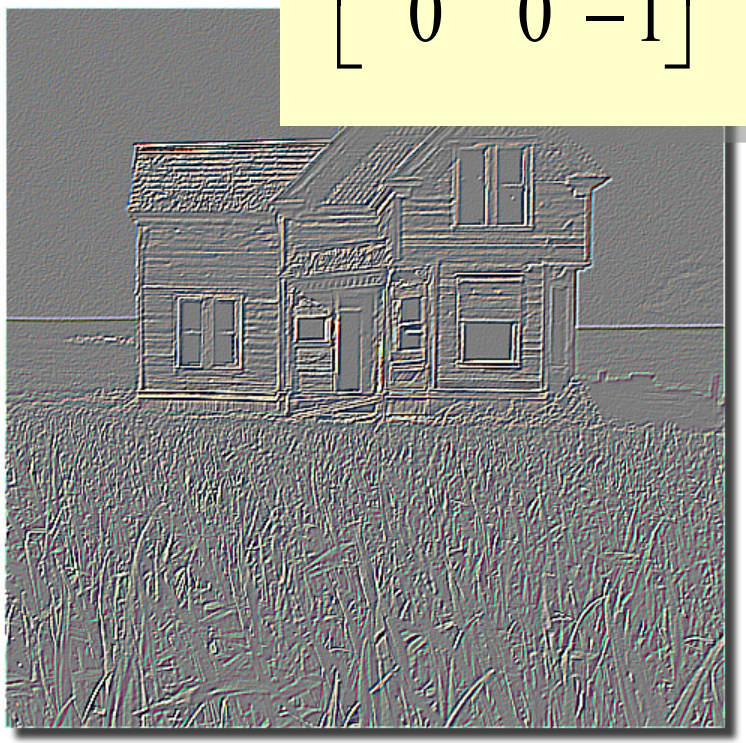


1st Derivative Filtering (cont…)

Convolution Examples:
Diagonal Difference

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

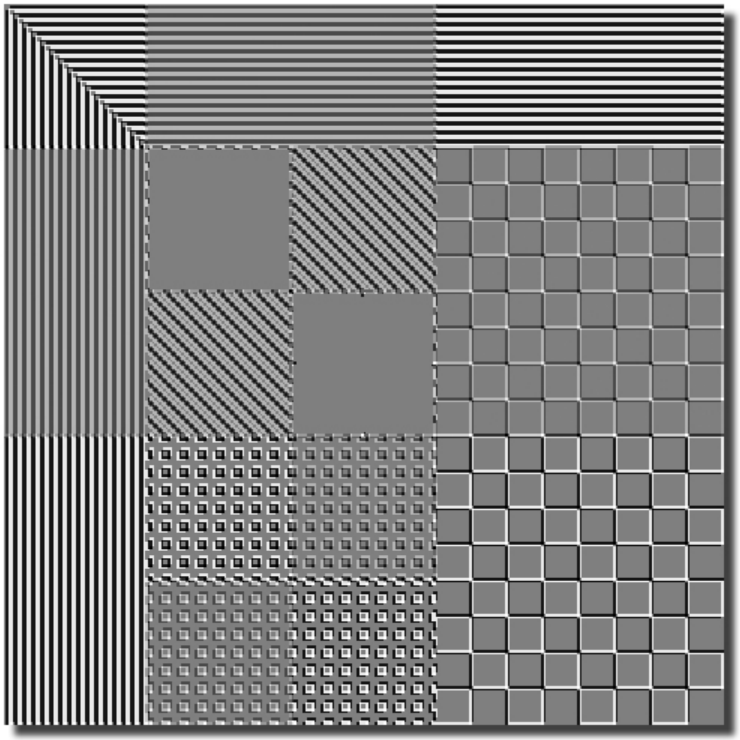


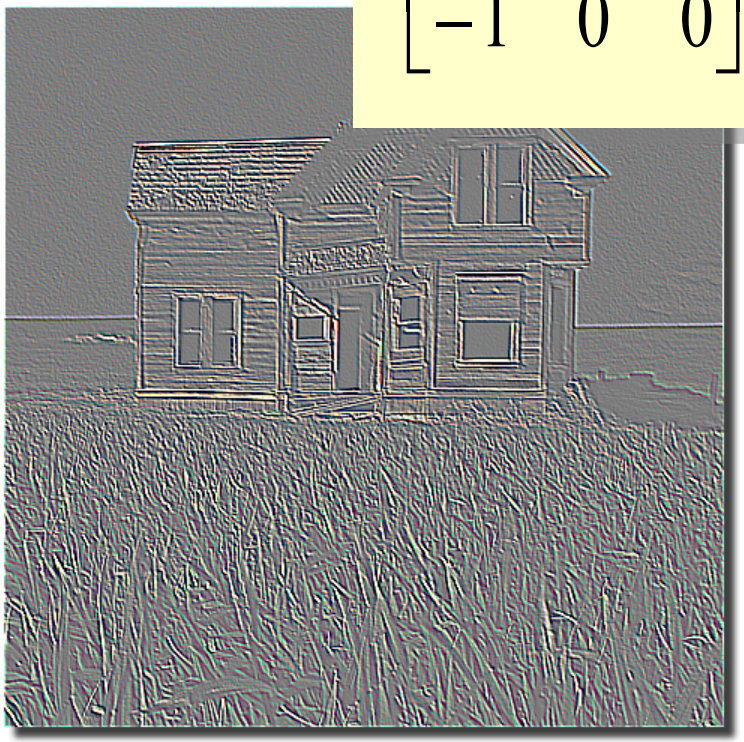


1st Derivative Filtering (cont…)

Convolution Examples:
Diagonal Difference

$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

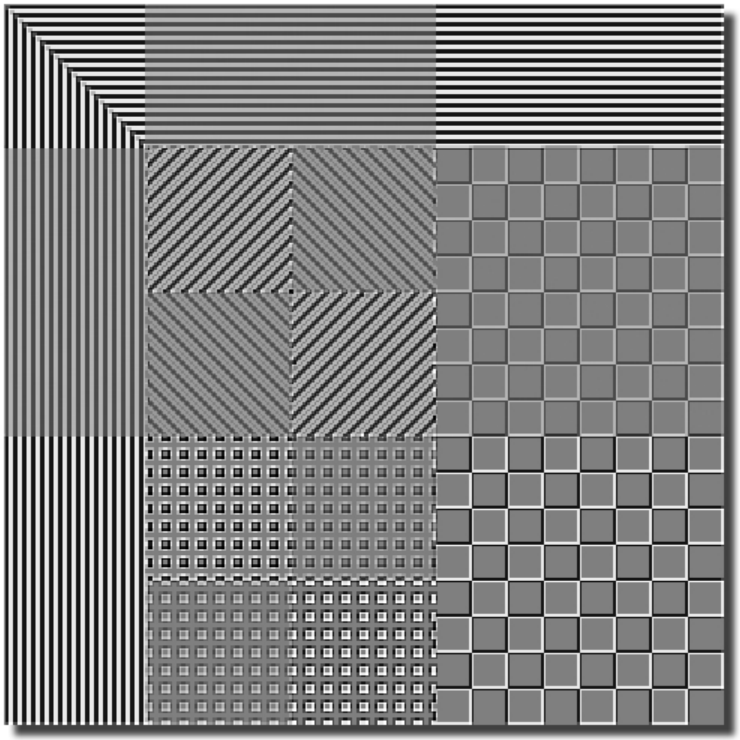


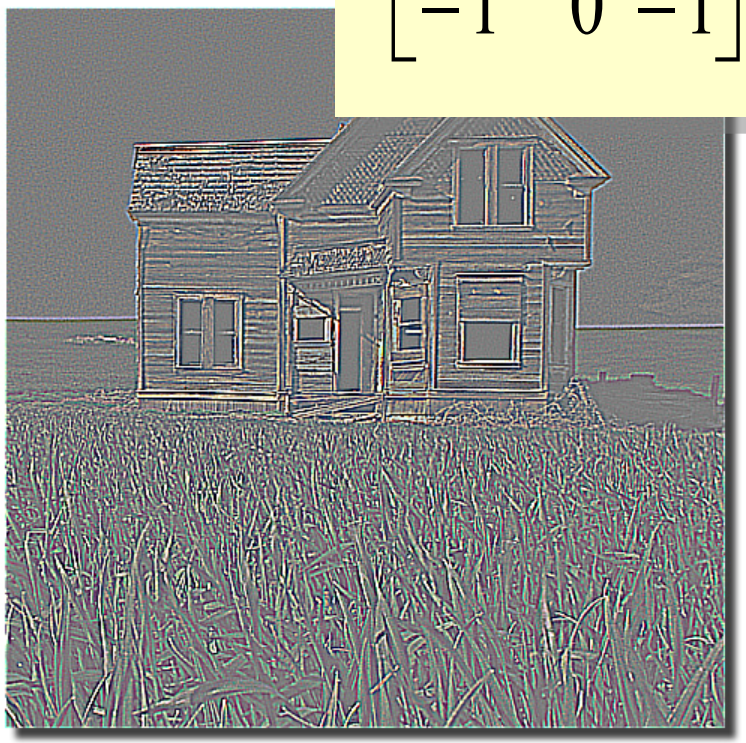


1st Derivative Filtering (cont…)

Convolution Examples:
D + D Difference

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

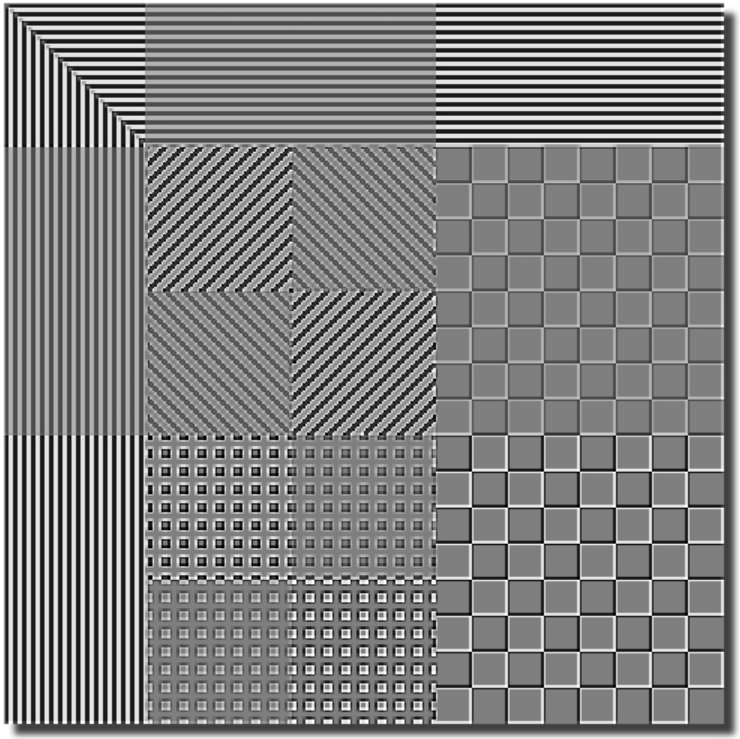


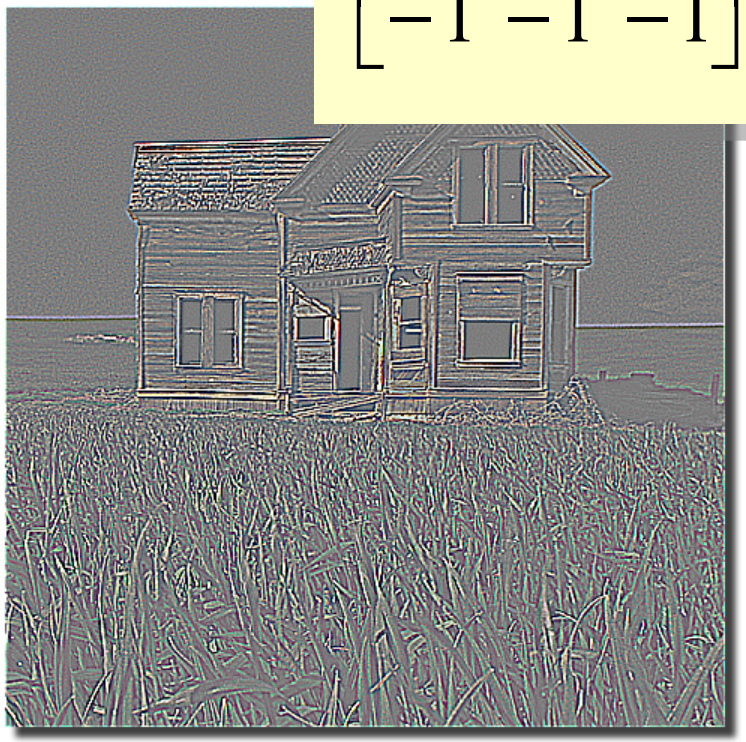


1st Derivative Filtering (cont…)

Convolution Examples:
H + V + D Diff.

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

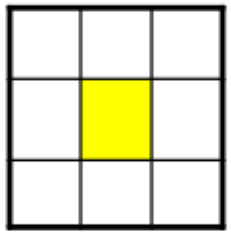




Vertical Difference

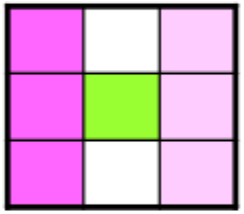


$$H = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

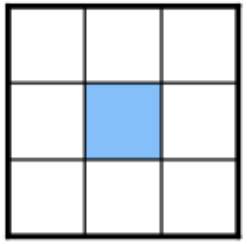




Horizontal Difference



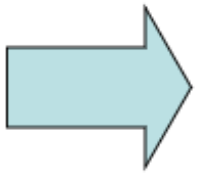
$$H = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$



$$H = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

1*1+2*2+1*3-1*3-2*0-1*8=-3

1	2	3	2	1
2	1	2	6	2
3	0	8	7	6
1	2	7	8	6
2	3	2	6	9



0	0	0	0	0
0	-3	-13	-20	0
0	-6	-13	-13	0
0	1	12	5	0
0	0	0	0	0

问题：计算结果中出现了小于零的像素值

方法1：整体加一个正整数，以保证所有的像素值均大于零。

这样做的结果是：可以获得类似浮雕的效果。

0	0	0	0	0
0	-3	-13	-20	0
0	-6	-13	-13	0
0	1	12	5	0
0	0	0	0	0

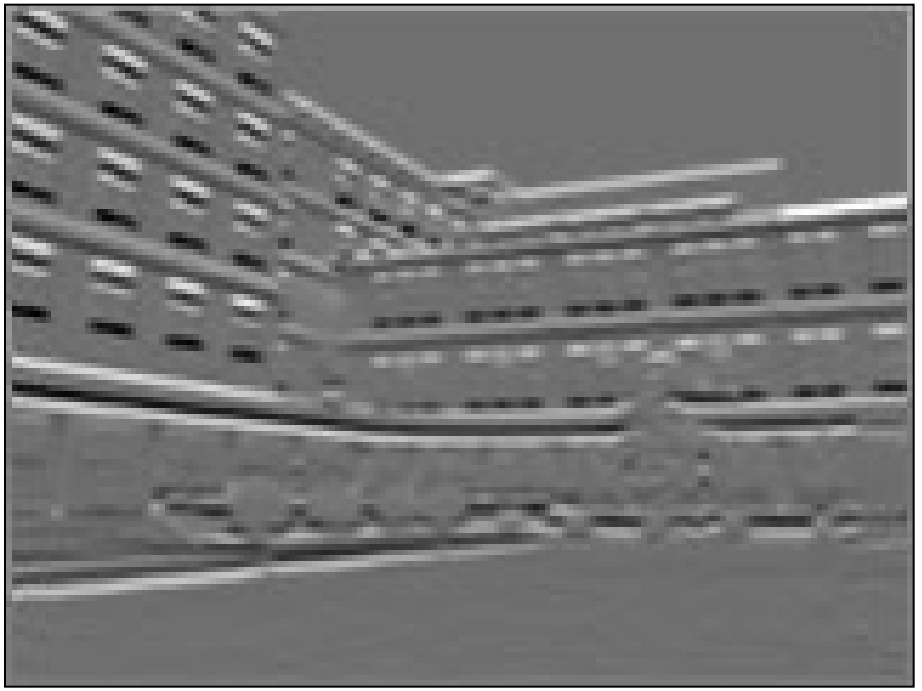
+20



20	20	2 0	20	20
20	17	7	0	20
20	14	7	7	20
20	21	32	25	20
20	2 0	20	2 0	20

Horizontal mask





方法2：将所有的像素值取绝对值。

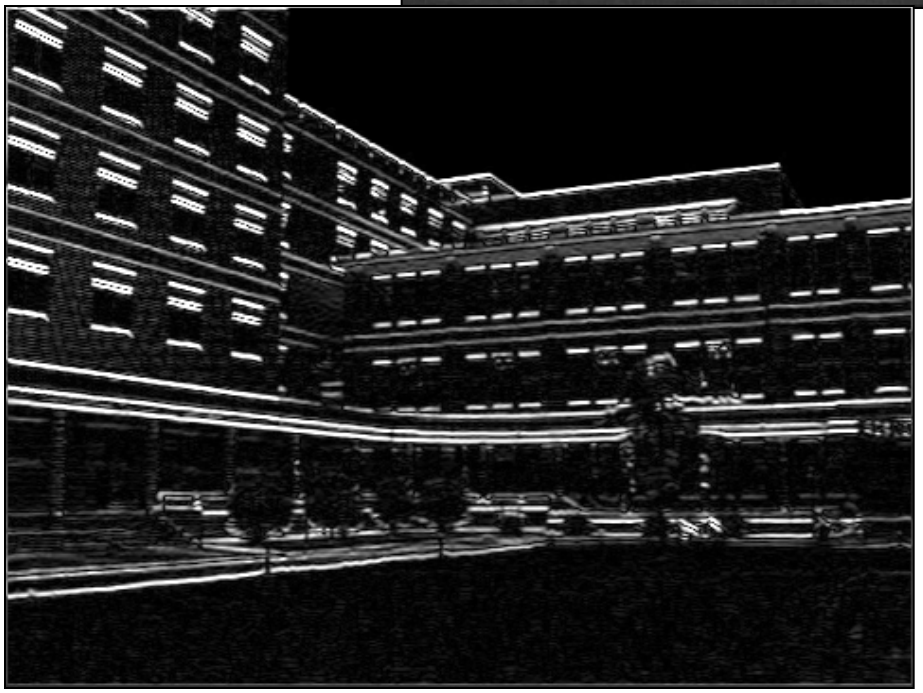
可获得比较清楚的边缘方向

0	0	0	0	0
0	-3	-13	-20	0
0	-6	-13	-13	0
0	1	12	5	0
0	0	0	0	0



0	0	0	0	0
0	3	13	20	0
0	6	13	13	0
0	1	12	5	0
0	0	0	0	0







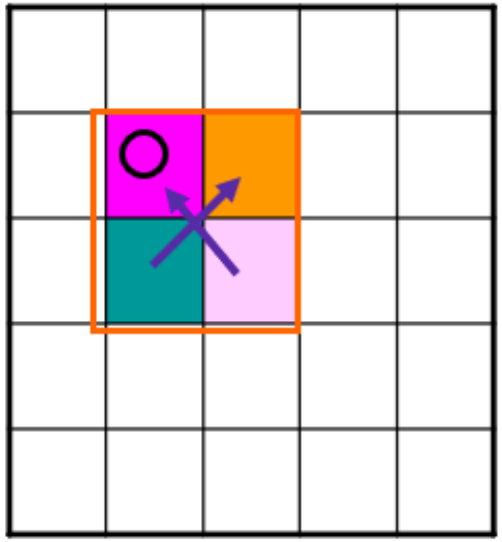
10.2.5 1st Derivative Filtering

- First derivatives sharpen algorithm with no direction
  - The Roberts cross-gradient operators
  - The Sobel operators
  - The Priwitt operators

1st Derivative Filtering (cont…)

- The Roberts cross-gradient operators

$$g(i,j) = |f(i+1,j+1) - f(i,j)| + |f(i+1,j) - f(i,j+1)|$$



Cross-gradient mask:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Algorithm is simple

1st Derivative Filtering (cont…)

- The Roberts cross-gradient operators





1st Derivative Filtering (cont…)

- The Roberts cross-gradient operators





1st Derivative Filtering (cont…)

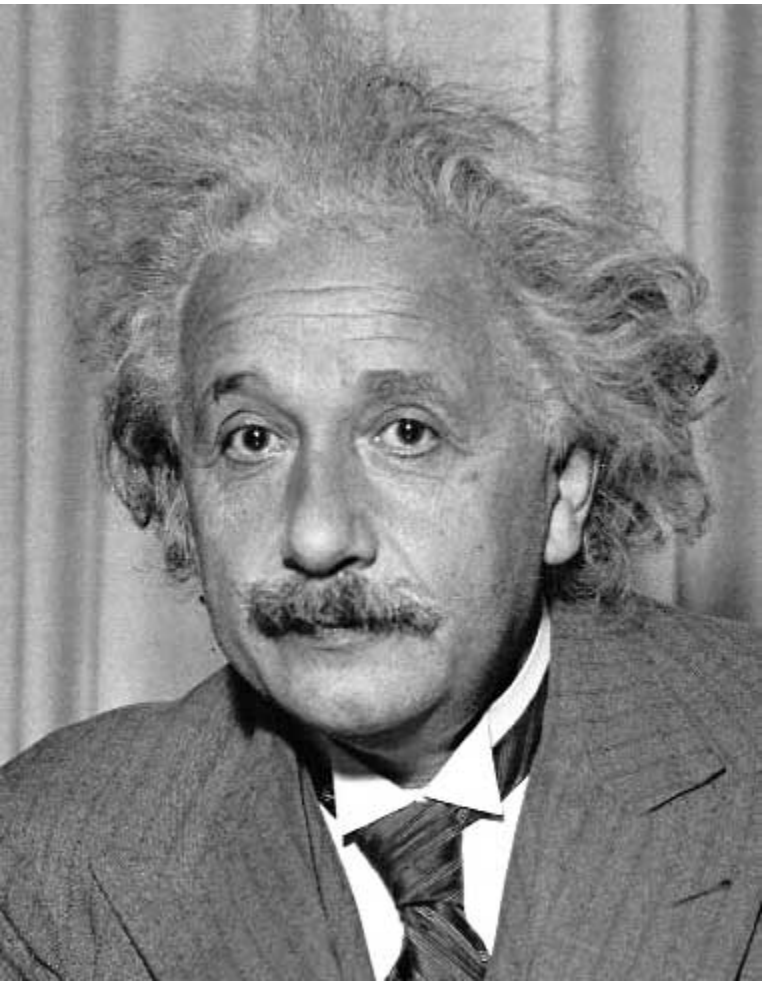
- The Sobel operators

$$g(i,j) = \{d_x^2(i,j) + d_y^2(i,j)\}^{\frac{1}{2}}$$

$$d_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \qquad d_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

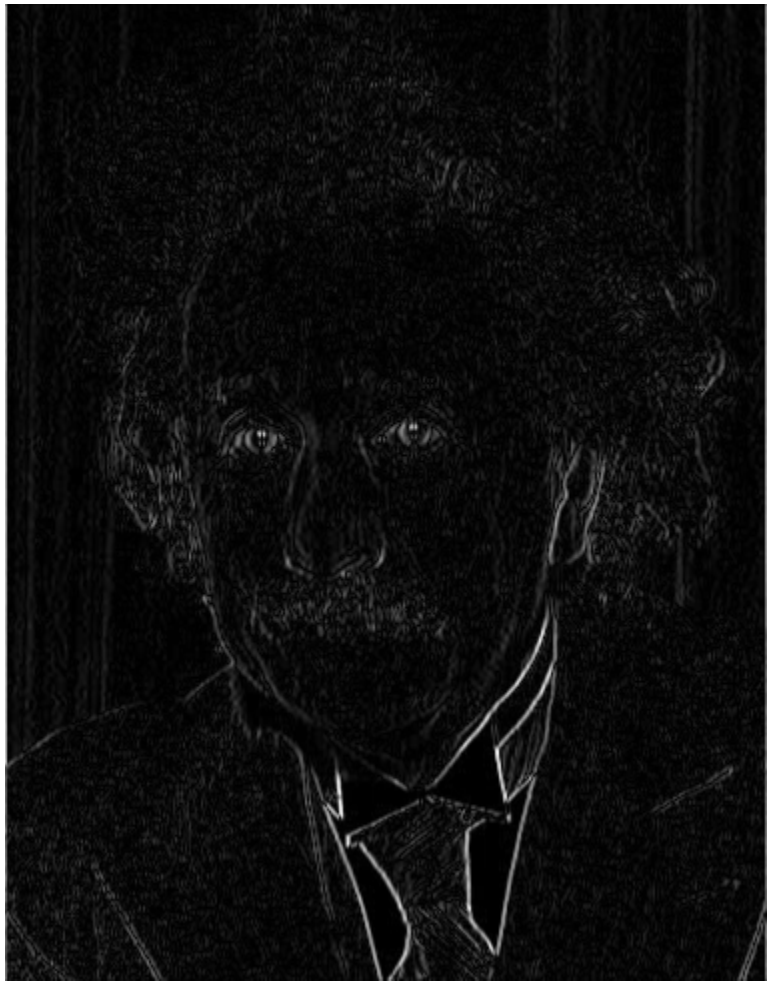
Get more edge information

Other filters （Sobel）



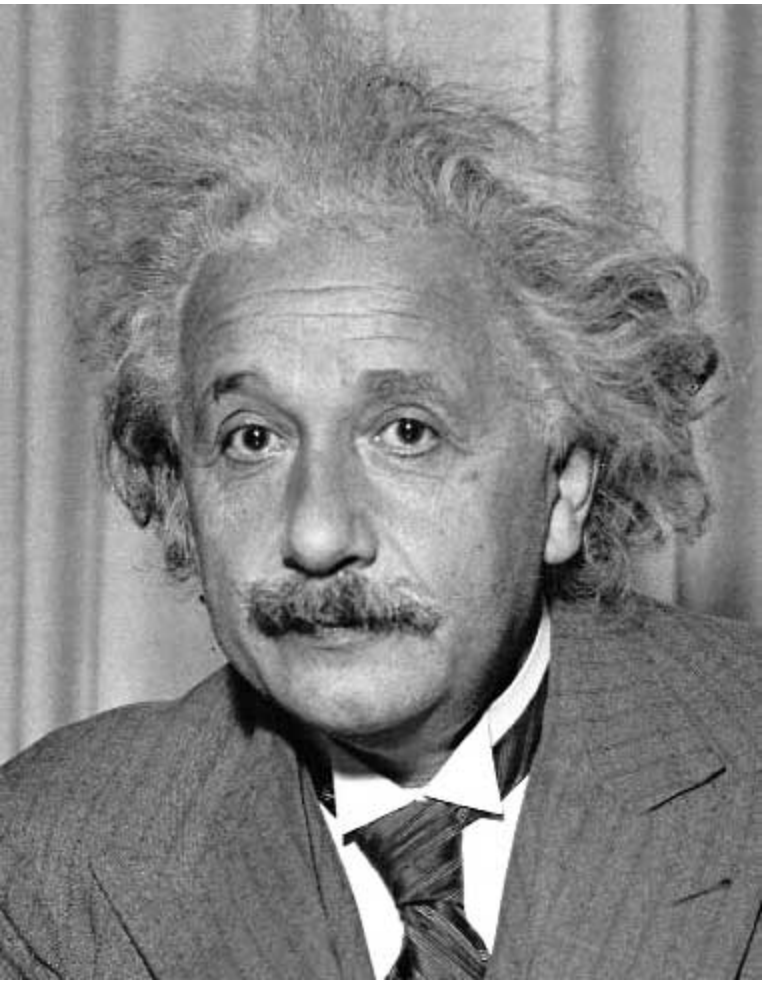
1	0	-1
2	0	-2
1	0	-1

Sobel



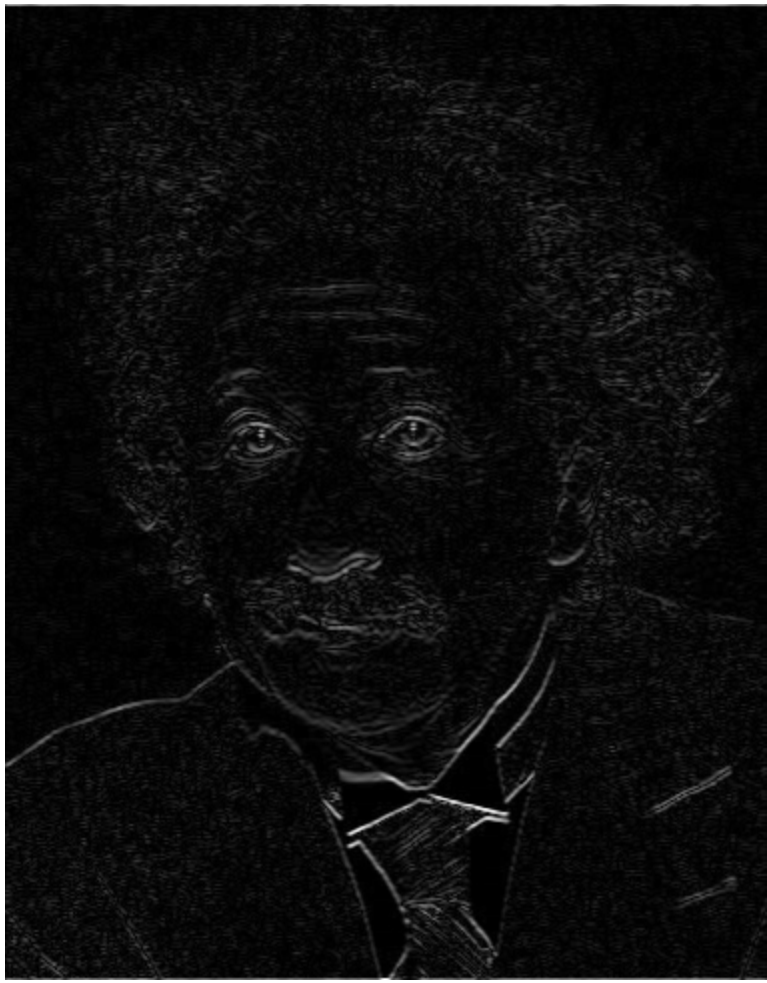
Vertical Edge
(absolute value)

Other filters （Sobel）



1	2	1
0	0	0
-1	-2	-1

Sobel



Horizontal Edge
(absolute value)

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

So these kernels have the effect of smoothing followed by gradient estimation

1st Derivative Filtering (cont…)





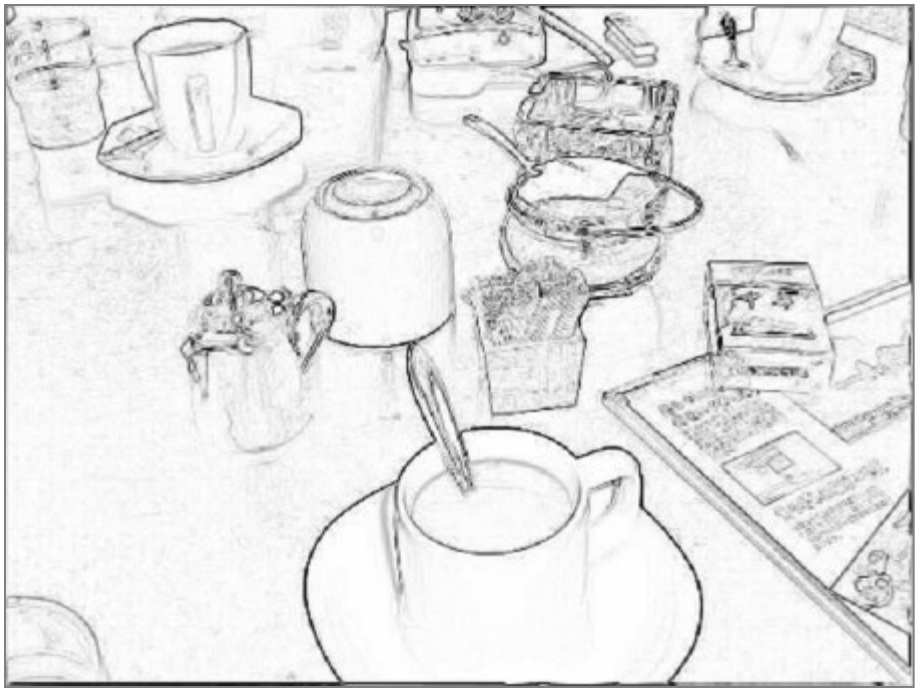
1st Derivative Filtering (cont…)

- The Priwitt operators

$$g(i,j) = \{d_x^2(i,j) + d_y^2(i,j)\}^{\frac{1}{2}}$$

$$d_x = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \qquad d_y = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

–The filter response is more sensitive to the edge orientation than the Sobel operator.



Roberts 交叉锐化



Sobel锐化

1st Derivative Filtering (cont…)



Sobel operator



Priwitt operator

特点：与Sobel相比， Priwitt有一定的抗干扰性。图像效果比较干净。

–The filter response is more sensitive to the edge orientation than the Sobel operator.



(a) Original



(b) Sobel



(c) Priwitt



(d) Roberts

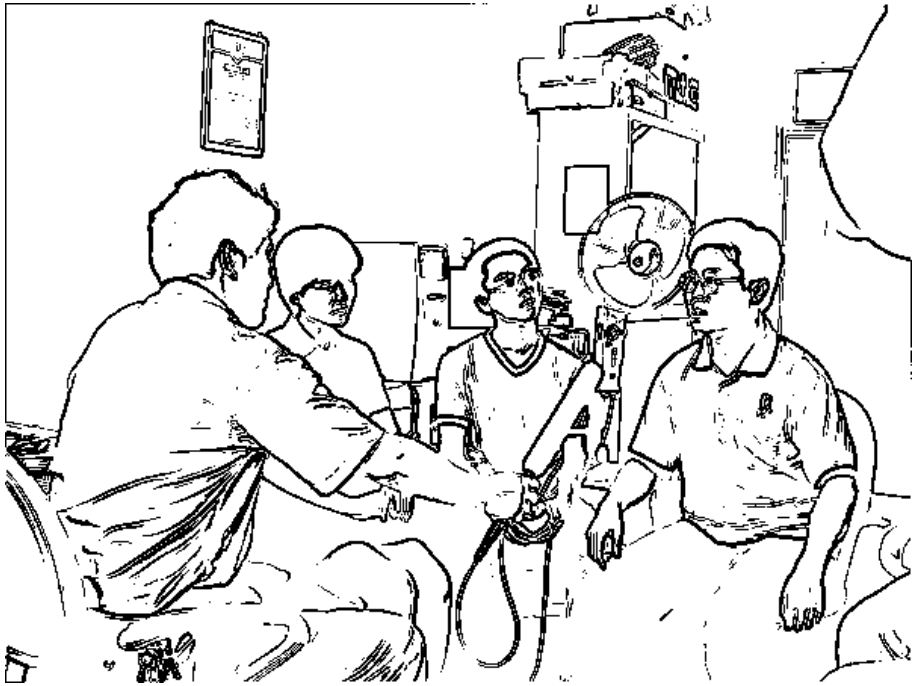


(e) Horizon

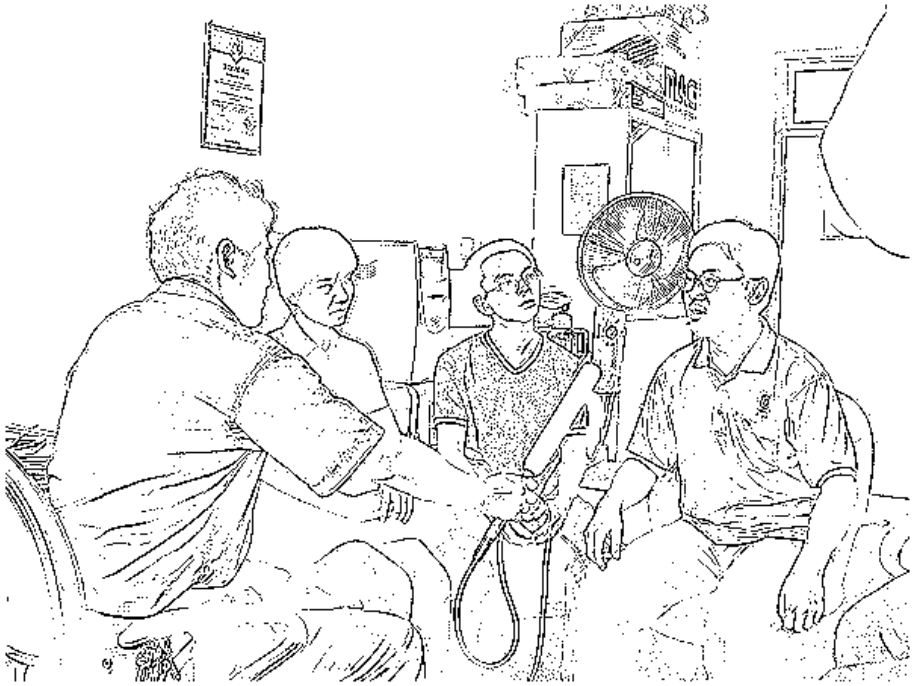


(f) Vertical

Sobel and Laplacian



Sobel锐化



Laplacian锐化

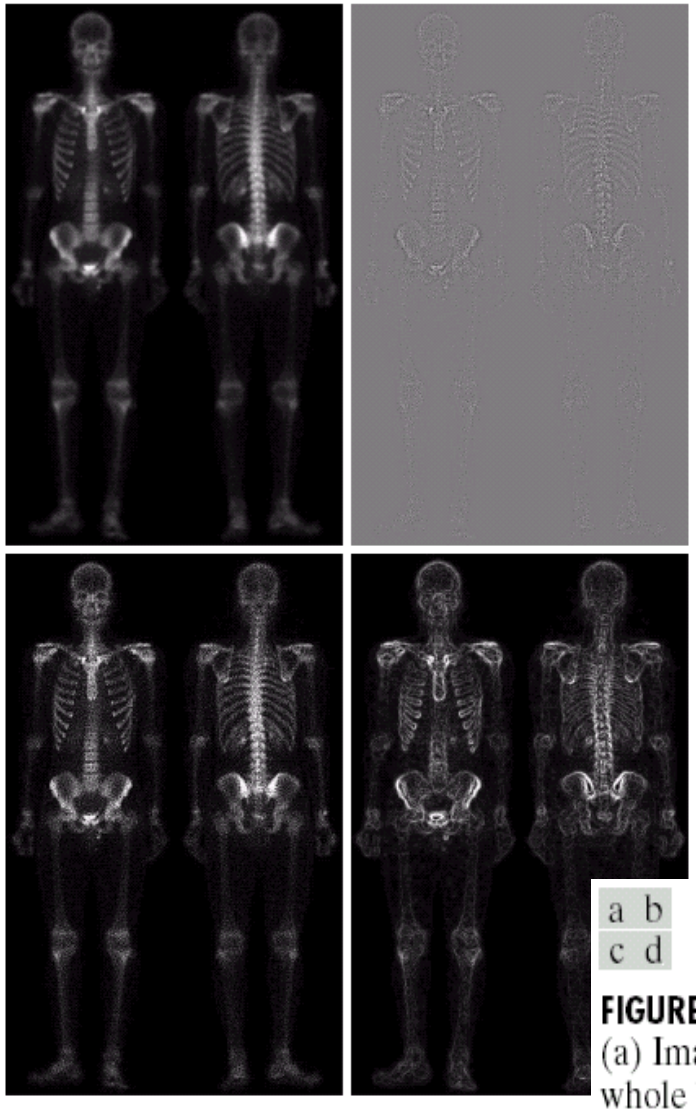
- Sobel算子获得的边界是比较粗略的边界，反映的边界信息较少，但是所反映的边界比较清晰；

- Laplacian算子获得的边界是比较细致的边界。反映的边界信息包括了许多的细节信息，但是所反映的边界不是太清晰。

Notes

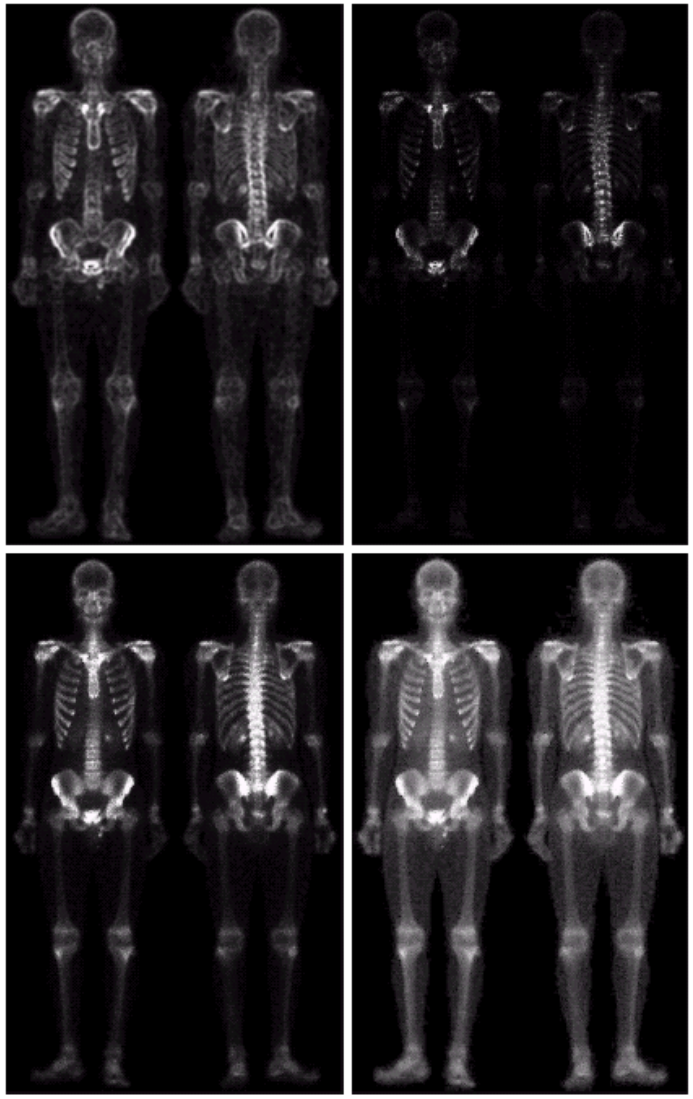
A single technique is not suitable for solving a real-life problem.

What process (image processing tools) we used, depends on the application.



a b
c d

FIGURE 3.46
(a) Image of whole body bone scan.
(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel of (a).



e f
g h

FIGURE 3.46
(Continued)
(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e).
(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

Summary

In this lecture we looked at:

- Sharpening filters
  - 1st derivative filters
  - 2nd derivative filters

单选题 25分

设置

1. Which of the following algorithms belongs to image sharpening:

A Low-pass filtering

B Weighted average

C High-pass filtering

D Median filtering

提交

单选题 25分

设置

2. The filter $[-1\ 1]^T$ mainly detects edges in the ( ) direction

A horizontal

B 45°

C vertical

D 135°

提交

多选题 50分

设置

3. Compare the characteristics of gradient and Laplacian operators

A Both operators are based on first derivative

B Gradient operator is based on first derivative; Laplacian is based on second derivative

C Edges are maximal points in gradient operators and are zero crossings in Laplace operators

D Both operators can extract edges and are sensitive to noise

提交